Data Collection and Preprocessing Phase

Date	20 June 2024
Team ID	739900
Project Title	Predicting Permanent Magnet Resistance Of Electronic Motor Using Machine Learning.
Maximum Marks	6 Marks

Data Exploration and Preprocessing Report :

Dataset variables will be statistically analyzed to identify patterns and outliers, with Python employed for preprocessing tasks like normalization and feature engineering. Data cleaning will address missing values and outliers, ensuring quality for subsequent analysis and modeling, and forming a strong foundation for insights and redictions.

Univariate Analysis
Section

Data Overview

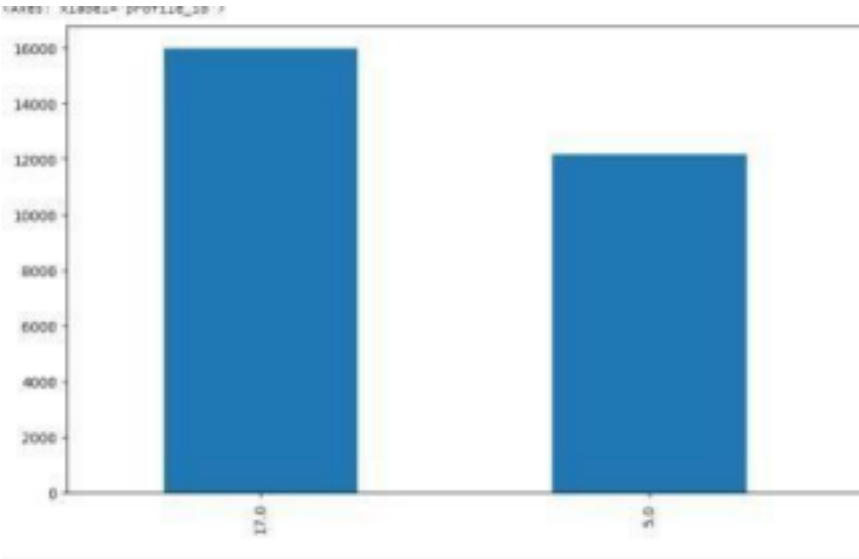
Description

Dimension:

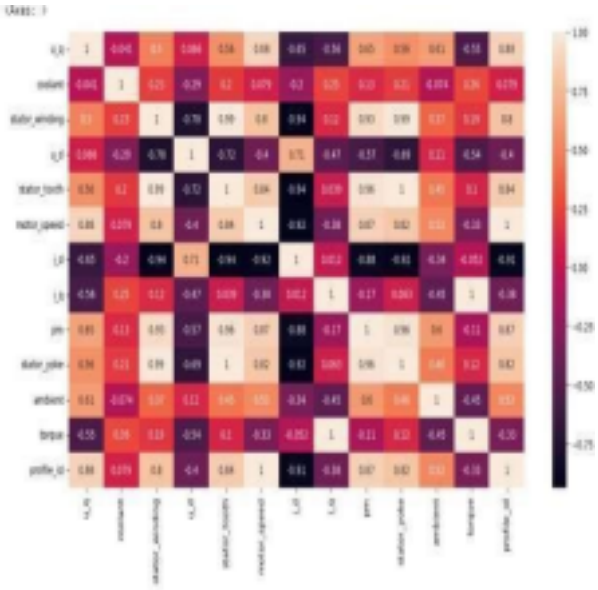
28115rows × 9 Columns Descriptive

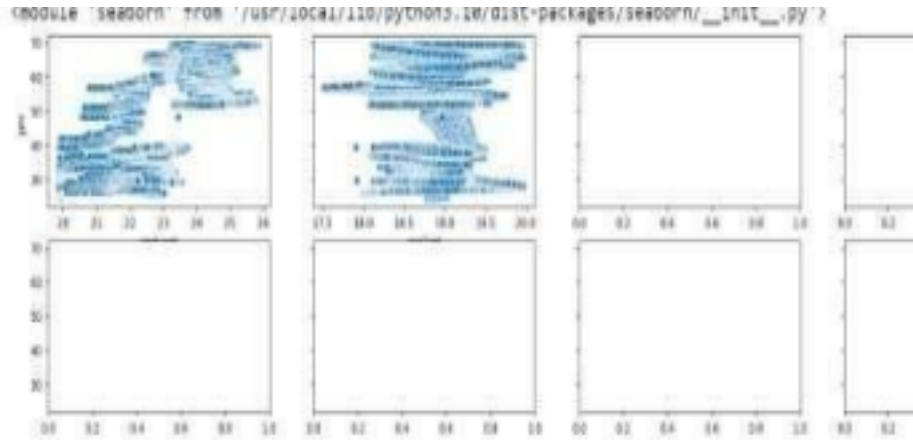
Descriptive statistics:

	u_q	coolant	u_d	motor_speed	i_d	i
0	-0.450682	18.805172	-0.350055	0.002866	0.004419	0.0003
1	-0.325737	18.818571	-0.305803	0.000257	0.000606	-0.0007
2	-0.440864	18.828770	-0.372503	0.002355	0.001290	0.0003
3	-0.327026	18.835567	-0.316199	0.006105	0.000026	0.0020
4	-0.471150	18.857033	-0.332272	0.003133	-0.064317	0.0371
...	...	...	...	...	...	
28110	49.997669	19.196394	-0.340213	999.987427	-1.997597	1.0971
28111	50.007954	19.171013	-0.329662	999.995911	-2.002546	1.0972
28112	50.028343	19.096292	-0.346269	999.986389	-2.000041	1.0977
28113	50.011482	19.034031	-0.323677	999.989929	-1.998187	1.0967
28114	50.003902	18.971241	-0.344793	NaN	NaN	N



Multivariate Analysis





Outliers Anomal

Data Preprocessing Code Screenshots

Loading Data
Handling Missing Data

```
data=pd.read_csv("/content/measures_v2.csv")
```

df.head()

Data Transformation
Feature Engineering
Save Processed Data

Attached the codes in final submission.